

# AI-Based Laptop Price Prediction System

## Viva Questions and Answers

**Q1. What is the title of your project?**

Ans: AI-Based Laptop Price Prediction System.

**Q2. What is the objective of this project?**

Ans: To predict laptop prices using machine learning algorithms.

**Q3. Which programming language did you use?**

Ans: Python.

**Q4. Which libraries did you use?**

Ans: Pandas, NumPy, Scikit-learn, Joblib, and Streamlit.

**Q5. Which dataset did you use?**

Ans: A laptop specifications and price dataset containing around 893 records.

**Q6. What is machine learning?**

Ans: Machine learning enables computers to learn patterns from data and make predictions.

**Q7. Which algorithm gave the best result?**

Ans: Random Forest Regressor.

**Q8. Why did you choose Random Forest?**

Ans: It handles non-linear data well and provides high accuracy.

**Q9. What is feature engineering?**

Ans: The process of creating useful features from raw data.

**Q10. What is one-hot encoding?**

Ans: A technique to convert categorical data into numerical format.

**Q11. What is data preprocessing?**

Ans: Cleaning and transforming raw data before model training.

**Q12. What is the train-test split used?**

Ans: 80% training and 20% testing.

**Q13. What evaluation metric did you use?**

Ans:  $R^2$  Score.

**Q14. What was your model accuracy?**

Ans: The Random Forest model achieved an  $R^2$  score of approximately 0.833.

**Q15. What is Streamlit?**

Ans: A Python framework for building interactive web applications.

**Q16. What are the input features?**

Ans: Brand, RAM, RAM Type, Storage, CPU, GPU, OS, Display Size, Resolution, and Warranty.

**Q17. What is the output of your system?**

Ans: Predicted laptop price.

**Q18. What are the advantages of your project?**

Ans: Fast prediction, user-friendly interface, and good accuracy.

**Q19. What are the limitations?**

Ans: Dataset size limitations and market price fluctuations.

**Q20. What is the future scope?**

Ans: Deep learning, cloud deployment, recommendation systems, and mobile apps.